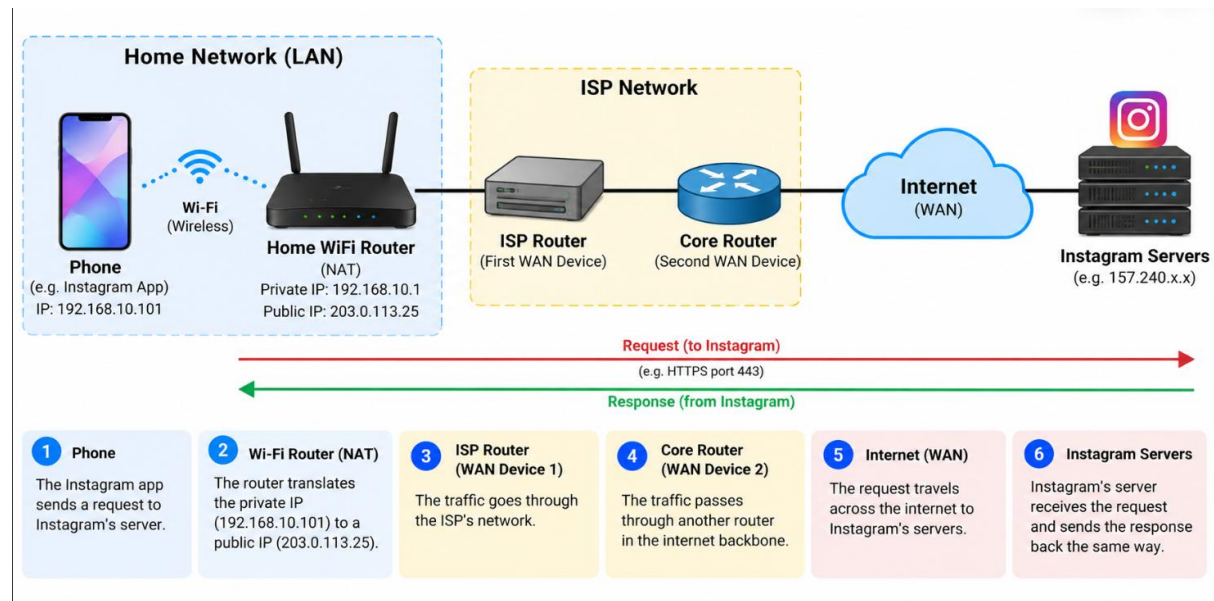


Wide Area Networks

Q – 1 : Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Solution : WAN Diagram



Explanation: The phone connects to the home Wi-Fi router, which sends the traffic through the ISP. The data then passes through multiple WAN routers before reaching Instagram's servers.

Q – 2 : Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Solution : **Comparison of WAN Technologies**

WAN Technology	Speed	Cost	Real-Life Use
MPLS	High	High	Large companies such as Zomato offices
Leased Line	High and dedicated	High	Banks, offices and enterprise networks
Broadband Internet	Medium to High	Low	Homes and small offices

Summary: MPLS and leased lines are generally used by organizations that need reliable dedicated connectivity, while broadband is commonly used for homes and small offices.

Q – 3 : Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Hint : Focus on how the VPN creates a private tunnel over the public WAN.

Solution : **VPN over a WAN**

A **VPN (Virtual Private Network)** creates an encrypted private tunnel through the public WAN/Internet.

Example:

Phone/Laptop → Coffee Shop Wi-Fi → Public WAN/Internet → VPN Server → Spotify/YouTube

The VPN encrypts the traffic between the device and VPN server, helping protect data from other users on the public Wi-Fi. The VPN then forwards the traffic to services such as Spotify or YouTube through the Internet.

Q – 4 : Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Solution : **WAN for Flipkart Warehouses**

Recommended: MPLS

I would recommend **MPLS** for connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi.

Reasons:

1. **Reliable connectivity:** MPLS provides managed and predictable connectivity between different locations.
2. **Suitable for large organizations:** It can connect multiple warehouses and offices while supporting business-critical applications and data communication.